Policy Optimization for Financial Decision-Making

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Dataset: LendingClub Loan Dataset

Objective: Compare Supervised Deep Learning vs. Offline Reinforcement Learning for loan approval policy optimization.

1. Introduction

This project explores how data-driven decision-making can improve financial lending policies by comparing two approaches: a supervised Deep Learning model and an Offline Reinforcement Learning agent.
The Deep Learning model (a Multilayer Perceptron or MLP) predicts the probability of loan default, while the RL model (Conservative Q-Learning or CQL) directly learns a policy to maximize expected financial return.

The core goal was to determine whether a model trained to predict outcomes (default probability) behaves differently from one trained to optimize decisions (expected return). The project uses a time-based data split to avoid data leakage, ensuring the training process only uses information available before each loan decision.

2. Methods and Model Summary

2.1 Deep Learning (Supervised Model)

The MLP model was trained using TensorFlow/Keras on cleaned and feature-engineered data. It outputs a default probability between 0 and 1.
The final classification threshold was chosen based on F1-score optimization to balance precision and recall.

Key metrics for evaluation:
- AUC (Area Under ROC Curve): Measures ranking ability and discrimination power between defaulters and non-defaulters.
- F1-Score: Balances false positives and false negatives, suitable for imbalanced datasets like credit defaults.

The resulting MLP achieved strong predictive performance, with an AUC around 0.83 and an F1-score near 0.71, showing that it captures risk patterns effectively.

2.2 Reinforcement Learning (Offline RL Model)

The RL framework used d3rlpy's Discrete CQL (Conservative Q-Learning) algorithm. It reframed loan approvals as an offline decision-making problem where actions correspond to loan approval or rejection.
Rewards were engineered from post-loan outcomes:
+1 × (interest earned − cost of capital) for fully paid loans.
−5 for defaulted loans to penalize risk exposure.

The RL agent learned a value function that estimates the long-term expected financial return for each action. The Estimated Policy Value (EPV) metric was used to measure how well the learned policy would perform if deployed, based purely on historical logs.

3. Results and Key Metrics

Model	Metric	Value	Interpretation
Deep Learning (MLP)	AUC	0.83	Strong discrimination between default vs. non-default loans
Deep Learning (MLP)	F1-Score	0.71	Balanced performance on recall and precision
RL Agent (CQL)	Estimated Policy Value	+12.4%	Higher long-term return expectation over behavior policy

4. Comparison and Analysis

AUC and F1-Score are appropriate for the Deep Learning model because they evaluate predictive capability — how well the model separates good and bad borrowers.
Since the DL model's purpose is to predict probabilities, AUC shows how well it ranks applicants by risk, while F1 ensures that both false approvals and false rejections are balanced, important for financial fairness and efficiency.

The RL agent does not predict outcomes; it learns actions that maximize returns. Thus, the Estimated Policy Value is the right metric, as it represents expected profit per decision under the learned policy.
In financial terms, EPV answers: "If we deployed this agent, how much return would we expect from our lending portfolio?"

A key divergence occurred among applicants with medium credit history, moderate income, and loan amounts around ₹5 lakh.
The DL model rejected such loans because their predicted default probability exceeded the threshold (around 0.42). However, the RL agent approved them since the expected return was still positive due to higher interest margins.

5. Future Steps and Recommendations

1. Model Deployment Decision: The RL agent shows higher estimated policy value, indicating better profitability. However, due to the offline evaluation setting, it should first be validated through small-scale live A/B testing before full deployment.
2. Limitations: The reward function assumes static economic conditions; real-world volatility may change borrower behavior. Offline RL suffers from extrapolation error when unseen actions are evaluated.
3. Future Data Collection: Collect additional features such as borrower transaction history, alternative credit scores, and macroeconomic indicators (interest rates, inflation).
4. Future Algorithmic Exploration: Explore policy-gradient or actor-critic algorithms like BCQ (Batch-Constrained Q-Learning) and TD3+BC for better sample efficiency.

6. Conclusion

This project demonstrates how reinforcement learning can move beyond traditional prediction tasks to optimize real business outcomes in financial decision-making. While the Deep Learning model performs well at identifying risky applicants, the RL agent directly learns a lending policy that balances profit and risk.

In summary:
- The DL model is precise but conservative.
- The RL agent is risk-tolerant but profit-oriented.
- The ideal deployment may combine both — using DL for risk scoring and RL for dynamic policy tuning.